LESSON PLAN · PYTHON (GR 6–8) BEGINNER: PYTHON GAME CODERS

Lesson 3: Numbers and Decisions

Module 1 | 60 minutes | int, maths, if/elif/else | Grades 6–10 · ages 11–15

Objective: Turn typed answers into numbers with int(), do maths in Python, and make decisions with if, elif and else.

Success criteria

- I can turn input into a number with int().
- I can do maths with + - * / // and %.
- I can use if, elif and else to make a decision.
- I can test my program with edge numbers.

Key vocabulary

- **int()**: Turns text like "42" into the number 42.
- **operator**: A maths symbol such as +, -, *, / or //.
- **condition**: A test that is True or False, like age < 12.
- **if / elif / else**: Run different code depending on conditions.
- **indent**: The four spaces that show which lines belong inside the if.
- **==**: Compares two values: is equal to?

You will need

- Projector with Kitabash (lesson video, Python Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: ticket_counter.py

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Human If-Statement**	Play Human If-Statement (unplugged): learners stand, raise hands or clap by the rules. Two learners invent rules.
5 min	**Hook and video**	Show the hook. Poll: “Is Rs 120 too costly? What decides?” Play the video with captions; pause on the 53 vs 8 scene and ask why.
10 min	**Live demo and build-along**	Run the live demo, then everyone builds the cinema ticket counter. Thumbs check: “Test with 12. Thumbs up if you get Adult.”
8 min	**Challenge, predict and match**	Ticket counter challenge, predict-the-output and symbol match. Quick competition: first to answer 17 // 5 and 17 % 5 in the chat.
15 min	**Project in breakout pairs**	Breakout pairs build the UPI shop calculator. One partner is the shopkeeper and tests with real prices; then swap.

6 min	**Debug race**	Race to fix “the pocket-money checker”: three bugs. First to explain all three wins.
7 min	**Show-and-tell**	Volunteers run their shop calculator while the class calls out prices. Did the discount rule work?
4 min	**Wrap-up and exit ticket**	Thumbs check on each success criterion. Exit ticket: Big or Small?

Unplugged warm-up

Human If-Statement (5 min game): the teacher reads out a rule such as “if your birthday is in a monsoon month (June to September), stand up; elif it is in winter, raise your hand; else, clap.” Learners act it out. Then a learner makes up the next rule using if, elif and else.

Teaching the idea

input() always returns text, so before we do maths we convert it with int(). Python can add, subtract, multiply and divide, and // gives whole-number answers that are perfect for rupees. A comparison like total >= 500 is either True or False. if runs its indented block only when the comparison is True; elif tries another test; else catches everything else. Python checks from the top and stops at the first True one.

Analogy: if/elif/else is like a railway signal at a junction: the train checks each signal in order and takes the first track that shows green.

Live demo: 1) Type a = input("First number: ") and b = input("Second number: ") then print(a + b). Run with 5 and 3 and show 53. Ask why. 2) Change it to print(int(a) + int(b)) and run again: 8. 3) Type price = 40, qty = 3, total = price * qty, print(total). Then try total // 2 and total % 7 and ask the class to predict. 4) Type runs = int(input("Runs? ")), then if runs >= 50: with an indented print("Half century!"), then else: print("Keep batting!"). Run with 67 and 12. 5) Remove the colon and run to show the SyntaxError, then remove the indent and run to show the IndentationError. Fix both.

Questions to ask

Ask	Listen for
Why does "5" + "3" give 53?	They are strings, so + joins them together.
What does 17 // 5 give? And 17 % 5?	3 and 2.
What is the difference between = and ==?	= stores a value; == compares two values.
If age is 12, which ticket does age < 12 give?	Not child, because 12 is not less than 12.
What must go at the end of an if line?	A colon.

Differentiation

Support
Give a printed if/elif/else skeleton with the colons and indents already there; learners fill in the conditions and messages.

Stretch
Add a “Diwali sale” rule: orders of Rs 1000 or more get 20% off, checked before the 10% rule with elif.

Common misconceptions

Misconception	How to fix it
input() already gives a number	It gives text. Wrap it in int() before doing maths.
= and == mean the same	= stores; == compares.
Python checks every elif even after one is True	Python stops at the first True condition.
Indentation is just for neatness	Indentation tells Python which lines belong inside the if.

Assessment

Use the Python Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What does this print: x = 7, then if x > 5: print("Big") else: print("Small")?

Homework
Use your shop calculator to price a real shopping list with someone at home. Add a rule of your own, such as free delivery over Rs 300.

Answer key — Kitabash activities

1. Python Lab: ticket counter (perform · python lab)

Goals checked: Age 8 gets a child ticket · Age 30 gets an adult ticket · Age 65 gets a senior ticket · Uses if, elif and else

One solution:

```
# Cinema ticket counter
age = int(input("How old are you? "))

if age < 12:
    print("Child ticket: Rs 100")
elif age < 60:
    print("Adult ticket: Rs 250")
else:
    print("Senior ticket: Rs 150")

print("Enjoy the film!")
```

2. Predict the output: half century? (perform · code output)

3. Match each symbol to what it means (perform · matching)

* → Multiply · // → Divide into a whole number · % → The remainder after dividing · == → Is equal to? · != → Is not equal to? · >= → Is more than or equal to?

4. Project: UPI shop calculator (perform · python lab)

Goals checked: Big order gets 10% off · Small order pays full price · Says how much more is needed · Uses if and else

Example program:

```
# UPI shop calculator
# Made by: Zoya

print("=== KITABASH CORNER SHOP ===")
price = int(input("Price of one item (Rs): "))
qty = int(input("How many? "))
pocket = int(input("Your pocket money (Rs): "))

total = price * qty
print(f"Total: Rs {total}")

# Discount rule: 10% off orders of Rs 500 or more
if total >= 500:
    discount = total * 10 // 100
    print(f"Big order! You save Rs {discount}")
else:
    discount = 0
    print("Spend Rs 500 or more to get 10% off.")

pay = total - discount
print(f"Pay Rs {pay}")

if pocket >= pay:
    change = pocket - pay
    print(f"You can pay with UPI! Left over: Rs {change}")
else:
    need = pay - pocket
    print(f"You need Rs {need} more.")
```

5. Debug it: the pocket-money checker (probe · python lab)

Goals checked: 150 means party time · 40 means save up

One solution:

```
money = int(input("Pocket money? "))
if money > 100:
    print("Party time!")
else:
    print("Save up!")
```

Answer key — worksheet

1. **Key words:** int() = Turns text like "42" into the number 42. · operator = A maths symbol such as +, -, *, / or //. · condition = A test that is True or False, like age < 12. · if / elif / else = Run different code depending on conditions.

2. **Plan your program:** Goals: Age 8 gets a child ticket; Age 30 gets an adult ticket; Age 65 gets a senior ticket; Uses if, elif and else. One solution: # Cinema ticket counter age = int(input("How old are you? ")) if age < 12: print("Child ticket: Rs 100") elif age < 60: print("Adult ticket: Rs 250") else: print("Senior ticket: Rs 150") print("Enjoy the film!")
3. **Predict the output: half century?:** A. Keep going! then Next ball...
4. **Match each symbol to what it means:** * → Multiply · // → Divide into a whole number · % → The remainder after dividing · == → Is equal to? · != → Is not equal to? · >= → Is more than or equal to?
5. **Plan your program:** Goals: 150 means party time; 40 means save up. One solution: money = int(input("Pocket money? ")) if money > 100: print("Party time!") else: print("Save up!")

Kitabash original curriculum.